EX:11 Network address translation

Aim:

To configure and examine network address translation using cisco packet tracer and verify communication between a private network and a public network.

Procedure:

1. Open cisco packet tracer.
2. Add a router, switch
3. Connect all devices using cables.
4. Configure the router's inside and outside interface.
5. Configure static NAT.
6. Assign IP address to the PC and server.
7. Test connectivity using ping and verify NAT translations.

Result:

Thus, the NAT was configured and verified successfully.

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successfully.

Wireshark · Packet 60 · wlp0s20f3

```
▾ Frame 60: Packet, 86 bytes on wire (688 bits), 86 bytes captur
    Section number: 1
  ▸ Interface id: 0 (wlp0s20f3)
    Encapsulation type: Ethernet (1)
    Arrival Time: Oct 14, 2026 08:59:17.646298401 IST
    UTC Arrival Time: Oct 14, 2026 03:29:17.646298401 UTC
    Epoch Arrival Time: 1791948557.646298401
    [Time shift for this packet: 0.000000000 seconds]
    [Time delta from previous captured frame: 5.992 microseconds
    [Time delta from previous displayed frame: 5.992 microsecond
    [Time since reference or first frame: 6.240677049 seconds]
    Frame Number: 60
    Frame Length: 86 bytes (688 bits)
    Capture Length: 86 bytes (688 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
    [Protocols in frame: eth:ethertype:ipv6:icmpv6]
    Character encoding: ASCII (0)
    [Coloring Rule Name: ICMP]
    [Coloring Rule String: icmp || icmpv6]
▸ Ethernet II, Src: Dell_47:06:da (cc:96:e5:47:06:da), Dst: IPv6
▸ Internet Protocol Version 6, Src: fe80::b81b:7b98:ea69:ecff, D
▾ Internet Control Message Protocol v6
    Type: Neighbor Solicitation (135)
    Code: 0
    Checksum: 0xf127 [correct]
    [Checksum Status: Good]
    Reserved: 00000000
    Target Address: fe80::1a3e:d659:51b1:4252
  ▸ ICMPv6 Option (Source link-layer address : cc:96:e5:47:06:da
```

```
0000  33 33 ff b1 42 52 cc 96  e5 47
0010  00 00 00 20 3a ff fe 80  00 00
0020  7b 98 ea 69 ec ff ff 02  00 00
0030  00 01 ff b1 42 52 87 00  f1 27
0040  00 00 00 00 00 00 1a 3e  d6 59
0050  cc 96 e5 47 06 da
```

No.: 60 · Time: 6.240677049 · Source: fe80::b81b:7b98:ea69:ecff · Destination: ff02... · Info: Neighbor Solicitation for fe80::1a3e:d659:51b1:4252 from cc:96:e5:47:06:da

☑ Show packet bytes Layout: Horizontal (Side-by-side) ✕ Close

Help